

GitHub Actions Field Guide

Empty secrets in fork PRs, silent cache misses and redundant billed runs.

4 fixes, each one a written note with diagrams and a small Python and Node.js script that finds the problem and repairs it. All 4 have a script in the public repo.

Before you run anything. Every script starts in a dry run: it reports what it would change and writes nothing. Read that output first, then set `DRY_RUN=false` when you are satisfied it has understood your data.

Scripts: github.com/allanninal/github-actions-fixes

Guides: allanninal.dev/ci

All 14 repos: github.com/allanninal

The fixes

1 a cache miss that is really a rate limit

The pipeline used to take four minutes and now takes eleven. Nothing failed. Nothing is red. The cache step reports Cache not found for input keys and the job installs everything from scratch, every time. The key has not changed and the cache still exists — the API declined to serve it, and actions/cache reports a decline the same way it reports an absence.

[actions_cache_audit.py](#)

<https://www.allanninal.dev/ci/cache-miss-is-really-a-rate-limit/>

<https://github.com/allanninal/github-actions-fixes/tree/main/cache-miss-is-really-a-rate-limit>

2 GITHUB_TOKEN is read-only and the error just says 403

The workflow builds fine and then dies on the last step with 403: Resource not accessible by integration. The token is right there in the environment. Nothing changed in the code. What changed, some time ago and for everyone, is the default:

GITHUB_TOKEN now starts read-only, and any step that pushes a commit, cuts a release or comments on an issue needs the permission granted explicitly.

[actions_permissions_audit.py](#)

<https://www.allanninal.dev/ci/github-token-is-read-only-by-default/>

<https://github.com/allanninal/github-actions-fixes/tree/main/github-token-is-read-only-by-default>

3 **three pushes run three full pipelines and you pay for all of them**

Somebody pushes a commit, spots a typo, pushes again, then fixes the lint error and pushes a third time. GitHub starts three full pipeline runs. Two of them are testing code that is already obsolete before they finish, and you are billed for every minute of all three. On a macOS runner, where minutes count at ten times the rate, those two wasted runs can cost more than the one you needed.

[actions_redundant_runs.py](#)

<https://www.allanninal.dev/ci/redundant-runs-on-rapid-pushes/>

<https://github.com/allanninal/github-actions-fixes/tree/main/redundant-runs-on-rapid-pushes>

4 **secrets are empty strings in fork pull requests**

The workflow passes on main and on every branch pushed by someone with write access. An outside contributor opens a pull request and the same workflow fails somewhere strange — a deploy step authenticating as nobody, an API call returning 401, a test asserting on a config value that is suddenly blank. GitHub did not refuse to give the job its secrets. It gave them as empty strings, and the job carried on.

[fork_pr_secret_audit.py](#)

<https://www.allanninal.dev/ci/secrets-are-empty-in-fork-pull-requests/>

<https://github.com/allanninal/github-actions-fixes/tree/main/secrets-are-empty-in-fork-pull-requests>
